Original article

Institution of MECHANICAL ENGINEERS

Journal of PROCESS MECHANICAL ENGINEERING

Proc IMechE Part E:
J Process Mechanical Engineering
1–12

DOI: 10.1177/09544089241275779
journals.sagepub.com/home/pie
Sage

Entropy optimization in ternary hybrid nanofluid flow over a convectively heated bidirectional stretching sheet with Lorentz forces and viscous dissipation: A Cattaneo–Christov heat flux model

Rajavath Narayana Naik[1], S Suneetha[1], KS Srinivasa Babu[2] and M Jayachandra Babu[3]

Abstract

Bidirectional stretching sheet models can represent surfaces in heat exchangers where fluids flow under continuous deformation. Ternary hybrid nanofluids could be employed in these systems to increase heat transfer rates between fluids in heat exchangers and improve the efficiency of energy conversion processes. In this work, we explore the novel application of a ternary hybrid nanofluid (water with titanium dioxide, cobalt ferrite, and magnesium oxide nanoparticles) for enhanced heat transfer in heat exchangers modeled by a bidirectional stretching sheet. This approach offers a potential advancement over traditional nanofluids (with one or two nanoparticles). Furthermore, we present a comprehensive analysis that incorporates ohmic heating, Cattaneo–Christov heat flux, thermal radiation, viscous dissipation, and irreversibility. The governing equations are transformed into a system of nonlinear ordinary differential equations using appropriate similarity transformations and then solved using the bvp4c solver, a MATLAB built-in function. This study's findings reveal that the Eckert number and radiation parameter increase fluid temperature, while the thermal relaxation parameter leads to a reduction in the temperature of the fluid. It is detected that an increase in magnetic field parameter and volume fraction of TiO_2 (ϕ_T) results in a decline of the skin friction factors in both directions. It is revealed that there is a reduction in the Nusselt number with the rise in Eckert number (Ecn), and the same number declines at a rate of 0.8252 when $0 \leq Ecn \leq 0.3$. It is noticed that the skin friction coefficient declines at a rate of 0.62179204 (in case of x-direction) and 0.621791816 (in case of y-direction), respectively, when the values of magnetic field parameter lie between 0 and 3.5. Furthermore, it is noticed that an upsurge in thermal relaxation parameter results in a fall in the temperature of the fluid.

Keywords

Nanofluid, entropy generation, grid independency, non-Fourier heat flux, stretching sheet, viscous dissipation

Date received: 19 March 2024; accepted: 8 July 2024

Introduction

A colloidal mixture of ordinary liquids with particles smaller than a nanometer in diameter is called a nanofluid. These particles are used to enhance ordinary liquids with low thermal conductivities' thermal properties. The most recent generations have reached different rates of thermal capabilities by utilizing a variety of state-of-the-art techniques to accelerate heat transfer rates. Enhancing heat conductivity is necessary to achieve this. Ultimately, there were several attempts to increase thermal conductivity by dispersing bigger solid thermally conductive components throughout the fluids. Numerous investigations on nanofluids have been carried out to satisfy the requirements of commercial uses. Scientists and experts in energy utilization may find that nanofluids meet their needs, although research is still ongoing to find a superior kind of fluid. Better nanofluid forms such as "hybrid nanofluid" with a higher thermal conductivity than nanofluid have emerged in response to these issues. These

[1]Department of Applied Mathematics, Yogi Vemana University, Kadapa, Andhra Pradesh, India
[2]Department of EM&H, S.R.K.R. Engineering College, Bhimavaram, Andhra Pradesh, India
[3]Department of Mathematics, Government Degree College, Rajampeta, Andhra Pradesh, India

Corresponding author:
KS Srinivasa Babu, Department of EM&H, S.R.K.R. Engineering College, Bhimavaram - 534204, Andhra Pradesh, India.
Email: kssb@srkrec.ac.in

particular nanoparticles have excellent thermal characteristics. A combination of several kinds of nanoparticles results in improved heat conductivity. In order to address problems in the real world, nanofluid is used in a range of industrial and nanotechnological applications, such as heat transfer systems, electronic device cooling, nuclear reactors, vehicle cooling, and vehicle thermal management. In addition, there are numerous other applications for magnetic nanofluids, such as cancer treatment, wound healing, artery opening, and magnetic resonance imaging. Over the past few years, owing to the said applications, plenty of works are found in literature. Medebber et al.[1] investigated the effect of a radial magnetic field on buoyancy-driven convection in a vertical annulus filled with Cu-water nanofluid. It is observed that the heat transfer feature of a nanofluid increases remarkably with the volume fraction of nanoparticles. Radouane et al.[2] performed a numerical investigation to analyze heat transfer of nanofluid within a two-dimensional enclosed system comprising a central heating cylinder. Jamshed et al.[3] inspected the dynamics of heat transfer in a power-law nanofluid flow through a porous surface with variable thermal conductivity, variable viscosity, and an applied transverse magnetic field. Wakif et al.[4] discussed the laminar flow of a nanofluid over an impermeable rotating disk with activation energy, thermophoresis, and chemical reaction parameters. It is detected that, compared to other control parameters, the activation energy parameter has a declining impact on the wall temperature. Shamshuddin et al.[5] explored the effect of magnetism on the bioconvection nanofluid flow through a stretching surface with thermal radiation parameter. Choudhary et al.[6] examined the thermal characteristics on the ternary hybrid nanofluid flow along a symmetrical stretching surface using the two categories of nanomaterials by using Kerosene as base fluid. A free convective motion which is a mixtures of Zirconium dioxide, magnesium oxide and CNT (as hybrid nanofluid) via an inclined two-phase channel saturated with porous medium is studied by Pavithra et al.[7] In both aiding and opposing flow cases, a ternary hybrid nanofluid radiative motion over an exponentially stretchable surface is addressed by Nagaraja et al.[8] Sarfraz et al.[9] examined the free convective flow of a ternary hybrid nanofluid using water as a base fluid via an inclined porous surface. With the help of Cattaneo–Christov heat flux, Yaseen et al.[10] discussed the bioconvection of ternary hybrid nanofluid motion in three different geometries. Prakash et al.[11] presented outcomes for EMHD ternary hybrid nanofluid over a wedge embedded in Darcy Forchheimer porous medium along with suction/injection. Entropy generation optimization in a ternary hybrid nanofluid via a curved stretching surface is another investigation carried out by Mahaboobtosi et al.[12] Along a linear and nonlinear stretching sheet, using a water- and engine oil–based ternary nanofluid, a numerical investigation is reported by Hasnain and Abid.[13] Madhukesh et al.[14] presented flow and heat transfer characteristics of a ternary hybrid nanofluid over a permeable inclined cylinder/plate. Mahmood et al.[15] discussed the numerical results for a ternary hybrid nanofluid over a stretching/shrinking curved surface along with suction and Lorentz force. The primary goal of the current effort is to use a ternary hybrid nanofluid to boost rate of heat transmission based on the previous works addressed by several authors.

Thermal radiation is the mechanism by which energy or heat is transferred by electromagnetic waves. Radiation parameter is important when the temperature difference between the ambient fluid and the boundary surfaces is large. In engineering and physics, radiative influences are significant. Completing activities requiring space technology and high temperatures requires careful evaluation of the effects of radiation heat transfer on various flows. Furthermore, radiation effects are critical to understanding heat transfer in the polymer sectors, where heat-regulating elements barely affect the quality of the finished product. Furthermore, the effects of radiation are pertinent on solar radiation, gas turbines, spacecraft, aircraft, and liquid metal fluids. Because of essentiality of thermal radiation effect, plenty of works have been found in past and recent literature. Few of such recent works are mentioned here. Guedri et al.[16] reported a theoretical study on unsteady radiative flow of micropolar nanofluid over a deformable sheet. Rafique et al.[17] have studied a three-dimensional flow of ethylene-glycol nanofluid due to varying viscosity and thermal radiation bidirectionally. Heat transmission in a MHD Maxwell nanofluid subject to thermal radiation effect over a stretchable surface is another similar study of Mukhtar et al.[18] Salawu et al.[19] discussed the efficiency of solar thermal radiation and energy optimization for a Prandtl–Eyring nanofluid. Ramesh et al.[20] examined the thermal radiation and thermophoretic particle deposition in the flow of hybrid CNTs along rotating sphere. A numerical discussion on MHD Casson nanofluid with Hall current and thermal radiation is another work reported by Suresh Kumar et al.[21] Dharmaiah et al.[22] have studied the motion of Jeffrey nanofluid over a wedge with nonlinear thermal radiation effect. Over a permeable moving surface, Lund et al.[23] presented the effects of thermal radiation and Joule heating due to the flow Casson of hybrid nanofluid.

In a variety of relevant circumstances, the traditional Fourier's heat conduction law[24] has proven to be the most effective model for describing the heat transmission mechanism. Despite this, it has a significant drawback in that it produces a temperature field parabolic energy equation, which is in conflict with the causality principle. In his well-known study, Cattaneo[25] proposed a successful modification of the Fourier's model by adding a crucial component of thermal relaxation time. It is observed that taking this into account results in a hyperbolic energy equation for the temperature field, which permits the movement of heat via the limited speed propagation of thermal waves. Excitement surrounds the practical applications of this type of heat transmission, which range from modelling to skin burn injuries from nanofluid flows. It should be mentioned that a number of materials have long thermal relaxation times, including sand

(21 seconds), biological tissues (1–100 seconds), and NaHCO3 (29 seconds). To maintain the material-invariant formulation, Christov[26] substituted the Oldroyd's upper-convected derivative for the time derivative in Maxwell-Cattaneo's model. The literature refers to this model as the Cattaneo–Christov heat flux model. Over the years, various reports have been addressed on this particular model by several investigators. Few of the recent attempts declares it. Ali et al.[27] discussed the mixed convective flow over a rotating disk with slip effects under the Cattaneo–Christov heat flux model. Numerical study due to Cattaneo–Christov model over a stretchable surface in porous media is another similar study of Khattak et al.[28] Using a finite element approach, Qureshi[29] reported the results of an EMHD nanofluid with the same model. Rehman et al.[30] presented the outcomes for a renovated Jeffery–Hamel flow problem along with said modelling. Along with this model, a three-dimensional coupled flow and heat transfer characteristics of a non-Newtonian nanofluid is another investigation by Gowda et al.[31] A comparative analysis is carried out by Fatima et al.[32] under the aspects of Cattaneo–Christov heat flux model in a nonlinear radiative ternary hybrid flow over a stretching surface. Saraswathy et al.[33] conducted a theoretical analysis on bio-convective motion of micripolar fluid with this model. Elshehabey et al.[34] discussed numerical simulation on the thermosolutal convective motion of a nanofluid with the help of artificial intelligence due to this modelling. Under the thermal radiation on Darcy–Forchheimer flow of Reiner–Philippoff fluid with the effect of Cattaneo–Christov heat flux model is another recent study of Rehman et al.[35]

The fact which gains the attention of researchers is that, physical properties of the fluid may vary significantly with temperature, especially viscosity and thermal conductivity. By decreasing the viscosity across the momentum boundary layer, rising temperatures cause an increase in transport phenomena, which in turn affects the rate of heat transfer at the wall. Thus, the temperature dependent viscosity of the fluid must be considered in order to forecast the flow and heat transfer rates. The heat produced by internal friction in lubricating fluids and the resulting increase in temperature have an impact on the fluid's viscosity, making it no longer reasonable to assume that the fluid's viscosity is constant. So, the researchers concentrated particularly on the behavior of flow and heat transfer along with changing viscosity with temperature as well as thermal conductivity in several numerical simulations. For instance, numerical analysis corresponding to an unsteady MHD dusty flow over a stretching sheet with varying viscosity and thermal conductivity is performed by Manjunatha and Gireesha.[36] On a vertical channel, combined effects of variable thermal conductivity and viscosity on a free convective motion of viscous fluid has been reported by Umavathi et al.[37] Ajibade and Tafida[38] conducted a theoretical analysis on the Couette flow due to varying thermal conductivity and varying viscosity effects over a vertical channel. Aziz et al.[39] described the effects of variable viscosity and thermal conductivity for a thermal cooling system over a rotating disk. Porgar et al.[40] discussed the effects of thermal conductivity and viscosity of nanofluids. Mandal and Pal[41] presented dual solutions of a hybrid nanofluid flow due to the impacts of variable viscosity and variable thermal conductivity over a permeable exponentially shrinking Riga surface. They also discussed the entropy generation and stability analysis.

Most studies on nanofluid flow over stretching sheets focus on a single stretching direction. Analyzing a bidirectional stretching sheet introduces a new geometric complexity that can influence the flow patterns and heat transfer behavior. While research on binary nanofluids is extensive, using a ternary hybrid nanofluid with three nanoparticle types for better heat transfer properties is relatively new and offers a potential performance improvement. Ternary hybrid nanofluids with their superior thermal conductivity can improve heat exchanger performance in power plants, refineries, and air conditioning systems. Cattaneo–Christov model provides a more accurate description of heat transfer compared to the traditional Fourier's law, especially for situations with rapid temperature changes. So, with these things in mind, in this work, we theoretically investigated the ternary hybrid nanofluid flow through a bidirectional stretching sheet with viscous dissipation, non-Fourier heat flux, thermal radiation, and entropy generation. Outcomes are displayed in the form of graphs and bar diagrams.

Formulation

In this work, a ternary hybrid nanofluid (water with titanium dioxide, cobalt ferrite, and magnesium oxide nanoparticles) flowing over a bidirectional stretching sheet is considered. The model incorporates ohmic heating, Cattaneo–Christov heat flux, irreversibility, thermal radiation, and viscous dissipation. The current investigation is based on the following assumptions:

1. Flow is induced by a bidirectional stretching surface at $z = 0$.
2. Refer to Table 1 for appropriate numerical estimates of the thermodynamic properties of the base fluid and nanoparticles used in this study.
3. The sheet is stretched in the direction of $x-$axis and $y-$axis, and $z-$axis is taken normal to it.

Table 1. The fluid and nanomaterial properties utilized in this study along with their respective numerical values (Alharbi et al.[44]).

S. no.	Properties	H_2O (f)	TiO_2 (ϕ_T)	$CoFe_2O_4$ (ϕ_C)	MgO (ϕ_M)
1	$\rho(kgm^{-3})$	997.1	4250	4907	3560
2	$k(Wm^{-1}K^{-1})$	0.613	8.9538	3.7	45
3	$C_p(Jkg^{-1}K^{-1})$	4179	686.2	700	955
4	$\sigma(Sm^{-1})$	0.0000055	2,380,000	5510000000	1.42×10^{-3}

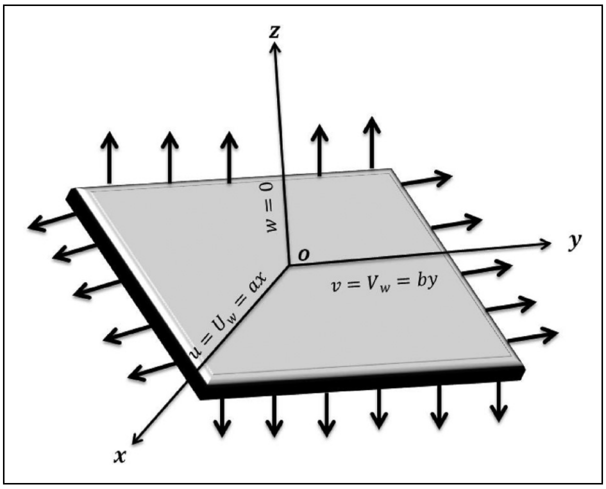

Figure 1. Flow geometry (Azam et al.[37]).

4. Magnetic field of strength B_0 is applied in the z-direction (see Figure 1).
5. Along the x-direction, elongating velocity of the sheet is $U_w = ax$, and the same along the y-direction is $V_w = by$.
6. The temperature on the sheet is T_w and temperature at a far distance from the sheet is T_∞.

$$
\begin{aligned}
&u\frac{\partial T}{\partial x}+v\frac{\partial T}{\partial y}+w\frac{\partial T}{\partial z}=\frac{1}{(\rho C_p)_{thnf}}\frac{\partial^2 T}{\partial z^2}\\
&-\chi\left[\begin{array}{c}
2\left(uv\frac{\partial^2 T}{\partial x\partial y}+vw\frac{\partial^2 T}{\partial y\partial z}+uw\frac{\partial^2 T}{\partial x\partial z}\right)+u^2\frac{\partial^2 T}{\partial x^2}+v^2\frac{\partial^2 T}{\partial y^2}+w^2\frac{\partial^2 T}{\partial z^2}\\
+\frac{\partial T}{\partial x}\left(u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}+w\frac{\partial u}{\partial z}\right)+\left(u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}+w\frac{\partial v}{\partial z}\right)\frac{\partial T}{\partial y}+\\
\frac{\partial T}{\partial z}\left(u\frac{\partial w}{\partial x}+v\frac{\partial w}{\partial y}+w\frac{\partial w}{\partial z}\right)
\end{array}\right]\\
&+\frac{1}{(\rho C_p)_{thnf}}\frac{16\sigma * T_\infty^3}{3k*}\frac{\partial^2 T}{\partial z^2}+\frac{1}{(\rho C_p)_{thnf}}\mu_{thnf}\left[\left(\frac{\partial u}{\partial z}\right)^2+\left(\frac{\partial v}{\partial z}\right)^2\right]\\
&+\frac{1}{(\rho C_p)_{thnf}}\sigma_{thnf}(u^2+v^2)
\end{aligned}
\tag{4}
$$

$$
\left.\begin{array}{c}
at\ z=0:\ u=U_w,\ v=V_w,\ w=0,\ -k_{thnf}\frac{\partial T}{\partial z}=h_f(T_w-T),\\
as\ z\to\infty:\ u\to 0,\ v\to 0,\ T\to T_\infty.
\end{array}\right\}
\tag{5}
$$

Thermal and physical properties of ternary hybrid nanofluid (THNF):

$$
\left.\begin{aligned}
\sigma_{nf}&=\sigma_f\times\frac{\sigma_M+2\sigma_f-2\sigma_f\phi_M+2\sigma_M\phi_M}{\sigma_M+2\sigma_f+\sigma_f\phi_M-\sigma_M\phi_M},\\
(\rho C_p)_{thnf}&=[(1-\phi_C)((1-\phi_M)(\rho C_p)_f+\phi_M(\rho C_p)_M)+\phi_C(\rho C_p)_C](1-\phi_T)+(\rho C_p)_T\phi_T,\\
\mu_{thnf}&=\frac{\mu_f}{(1-\phi_T)^{2.5}(1-\phi_C)^{2.5}(1-\phi_M)^{2.5}},\\
\sigma_{hnf}&=\sigma_{nf}\times\frac{\sigma_C+2\sigma_{nf}-2\sigma_{nf}\phi_C+2\sigma_T\phi_C}{\sigma_C+2\sigma_{nf}+\sigma_{nf}\phi_C-\sigma_C\phi_C},\\
\rho_{thnf}&=[(1-\phi_C)((1-\phi_M)\rho_f+\phi_M\rho_M)+\phi_C\rho_C](1-\phi_T)+\rho_T\phi_T,\\
\sigma_{thnf}&=\sigma_{hnf}\times\frac{\sigma_T+2\sigma_{hnf}-2\sigma_{hnf}\phi_T+2\sigma_T\phi_T}{\sigma_T+2\sigma_{hnf}+\sigma_{hnf}\phi_T-\sigma_T\phi_T},
\end{aligned}\right\}
$$

7. Induced magnetic field impact is disregarded in this work.

Here are the conditions (prerequisites) and basic equations needed for this study, based on these presumptions: (Azam et al.[42] and Ali et al.[43]):

$$
\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}=0 \tag{1}
$$

$$
u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}+w\frac{\partial u}{\partial z}=\frac{\mu_{thnf}}{\rho_{thnf}}\frac{\partial^2 u}{\partial z^2}-\frac{\sigma_{thnf}B_0^2}{\rho_{thnf}}u \tag{2}
$$

$$
u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}+w\frac{\partial v}{\partial z}=\frac{1}{\rho_{thnf}}\frac{\partial^2 v}{\partial z^2}-\frac{\sigma_{thnf}B_0^2}{\rho_{thnf}}v \tag{3}
$$

$$
\left.\begin{aligned}
k_{nf}&=k_f\times\frac{k_M+2k_f-2k_f\phi_M+2k_M\phi_M}{k_M+2k_f+k_f\phi_M-k_M\phi_M},\\
k_{hnf}&=k_{nf}\times\frac{k_C+2k_{nf}-2k_{nf}\phi_C+2k_C\phi_C}{k_C+2k_{nf}+k_{nf}\phi_C-k_C\phi_C},\\
k_{thnf}&=k_{hnf}\times\frac{k_T+2k_{hnf}-2k_{hnf}\phi_T+2k_T\phi_T}{k_T+2k_{hnf}+k_{hnf}\phi_T-k_T\phi_T}.
\end{aligned}\right\}
$$

Liu and Andersson[45] provided the succeeding similarity transmutations for converting governing equations:

$$
\left.\begin{array}{c}
\eta=z\sqrt{\frac{U_w}{\upsilon x}},\ u=axf',\quad w=-\sqrt{a\upsilon}(f(\eta)+g(\eta)),\ v=ayg',\\
\theta(\eta)=\frac{T-T_\infty}{T_w-T_\infty}.
\end{array}\right\}
\tag{6}
$$

By leveraging (6), the continuity Equation (1) seamlessly aligns, allowing for a straightforward fulfilment. Equations (2)–(4) can then be adeptly adjusted by incorporating Equation (5) in the subsequent process:

$$\frac{1}{S_1S_2}f''' + ff'' + gf'' - f'^2 - \frac{S_3}{S_1}Mnf' = 0 \tag{7}$$

$$\frac{1}{S_1S_2}g''' + gg'' + fg'' - g'^2 - \frac{S_3}{S_1}Mng' = 0 \tag{8}$$

$$\begin{aligned}&(S_4 - S_5 \Pr\Gamma(f+g)^2 + Rd)\frac{\theta''}{S_5\Pr(f+g)(f'+g')}\\ &+\frac{1}{S_2S_5}Ecn(f''^2 + \alpha^2 g''^2) + \frac{S_3}{S_2}MnEcn(f'^2 + \alpha^2 g'^2)\\ &= 0\end{aligned} \tag{9}$$

$$\left.\begin{array}{l}\text{at } \eta = 0{:}f = g = 0, \quad f' = 1, \quad g' = \lambda, \quad \theta' = -Bi(1-\theta),\\ \text{as } \eta \to \infty{:}f' \to 0, \quad g' \to 0, \quad \theta \to 0.\end{array}\right\} \tag{10}$$

Note that

$$\left.\begin{aligned}
S_1 &= (1-\phi_M)\left((1-\phi_C)\left[(1-\phi_T) + \phi_T\frac{\rho_T}{\rho_f}\right] + \phi_C\frac{\rho_C}{\rho_f}\right)\\
&\quad +\phi_M\frac{\rho_M}{\rho_f},\\
S_{311} &= \frac{k_T + 2k_f - 2\phi_T(k_f - k_T)}{k_T + 2k_f + \phi_T(k_f - k_T)},\\
S_{411} &= \frac{\sigma_T + 2k_f - 2\phi_T(\sigma_f - \sigma_T)}{\sigma_T + 2\sigma_f + \phi_T(\sigma_f - \sigma_T)},\\
S_3 &= \frac{k_M + 2S_{31}k_f - 2\phi_M(S_{31}k_f - k_M)}{k_M + 2S_{31}k_f + \phi_M(S_{31}k_f - k_M)}\times S_{31},\\
S_{41} &= \frac{\sigma_C + 2S_{411}\sigma_f - 2\phi_C(S_{411}\sigma_f - \sigma_C)}{\sigma_C + 2S_{411}\sigma_f + \phi_C(S_{411}\sigma_f - \sigma_C)}\times S_{411},\\
S_5 &= (1-\phi_M)\left((1-\phi_C)\left[(1-\phi_T)+\phi_T\frac{(\rho C_p)_T}{(\rho C_p)_f}\right] + \phi_C\frac{(\rho C_p)_C}{(\rho C_p)_f}\right)\\
&\quad +\phi_M\frac{(\rho C_p)_M}{(\rho C_p)_f},\\
S_{31} &= \frac{k_C + 2S_{311}k_f - 2\phi_C(S_{311}k_f - k_C)}{k_C + 2S_{311}k_f + \phi_C(S_{311}k_f - k_C)}\times S_{311},\\
S_4 &= \frac{\sigma_M + 2S_{41}\sigma_f - 2\phi_M(S_{41}\sigma_f - \sigma_M)}{\sigma_M + 2S_{41}\sigma_f + \phi_M(S_{41}\sigma_f - \sigma_M)}\times S_{41},\\
S_2 &= (1-\phi_T)^{2.5}(1-\phi_C)^{2.5}(1-\phi_M)^{2.5}.
\end{aligned}\right\}$$

and

$$\left.\begin{aligned}
&\Pr = \frac{\mu C_p}{k}, Rd = \frac{16}{3}\frac{\sigma * T_\infty^3}{k_f k*}, \lambda = \frac{b}{a}, \Gamma = a\chi, \alpha = \frac{V_w}{U_w},\\
&Mn = \frac{\sigma_f B_0^2}{\rho_f a}, Bi = -\frac{h_f}{k_f\sqrt{\frac{a}{\upsilon}}}, Ecn = \frac{U_w^2}{C_p(T_w - T_\infty)}.
\end{aligned}\right\}$$

In this work, we assume the condition (without losing generality) $(f+g)(0) = 0$ as $f(0) = g(0) = 0$, and the value of λ lies between 0 and 1.

Friction factor and transfer rate (heat) are defined as:

$$Cf_0 = \left.\frac{\tau_{zx}}{\frac{1}{2}\rho_f U_w^2}\right|_{z=0}, Cf_1 = \left.\frac{\tau_{zy}}{\frac{1}{2}\rho_f V_w^2}\right|_{z=0}, \tag{11}$$

where

$$\begin{aligned}&\tau_{zx} = \mu_{hnf}\frac{\partial u}{\partial z}, \tau_{zy} = \mu_{hnf}\frac{\partial v}{\partial z},\\ &q_w = -\left(k_{hnf} + \frac{16}{3}\frac{\sigma * T_\infty^3}{k*}\right)\frac{\partial T}{\partial z}\end{aligned}.$$

Terms in (11) can be recast using (6) as:

$$\begin{aligned}(Re_x)^{0.5}Cf_0 &= \frac{2}{S_2}f''(0), (Re_y)^{0.5}Cf_1\\ &= \frac{2}{S_2}\frac{1}{\lambda^{\frac{3}{2}}}g''(0), (Re_x)^{-0.5}Nu\\ &= -(S_4 + Rd)\theta'(0), Re_x = \frac{xU_w}{\upsilon}, Re_y\\ &= \frac{yV_w}{\upsilon}.\end{aligned} \tag{12}$$

Entropy generation and Bejan number

Entropy generation in fluid dynamics pertains to the inevitable escalation in the chaos or unpredictability present in a fluid system. Entropy consistently exhibits an upward tendency in practical situations, unlike certain other qualities that may vary. This occurrence is intricately associated with the Second Law of Thermodynamics, which dictates that the overall entropy of a closed system inevitably rises as time progresses. Reducing entropy production frequently results in enhancing energy efficiency. By delving into this subject, engineers can enhance their ability to create more effective pumps, turbines, heat exchangers, and other fluid systems.

The dimensional expression for calculating the total irreversibility in the current flow problem is articulated by the following formula:

$$\begin{aligned}S_G = &\frac{k_f}{T_\infty^2}\left(\frac{k_{thnf}}{k_f} + \frac{4}{3}\frac{4\sigma * T_\infty^3}{k_f k*}\right)\\ &\times\left(\frac{\partial T}{\partial z}\right)^2 + \frac{\mu_{thnf}}{T_\infty}\left[\left(\frac{\partial u}{\partial z}\right)^2 + \left(\frac{\partial v}{\partial z}\right)^2\right]\\ &+ (u^2 + v^2)\frac{\sigma_{thnf}B_0^2}{T_\infty}.\end{aligned} \tag{13}$$

Through the utilization of (6), Equation (13) can be rephrased as follows:

$$\begin{aligned}EGN = &\alpha_1(S_4 + Rd)\theta'^2 + \frac{Br}{S_2}\left(f''^2 + \alpha^2 g''^2\right)\\ &+ S_3 Br\left(f'^2 + \alpha^2 g'^2\right)Mn,\end{aligned} \tag{14}$$

where

$$EGN = \frac{\upsilon T_\infty S_G}{a(T_w - T_\infty)k_f},\ Br = \frac{\mu U_w^2}{k(T_w - T_\infty)},\ \alpha_1 = \frac{T_w - T_\infty}{T_\infty}.\Bigg\}$$

The Bejan number can be determined by use this mathematical expression:

$$BJ = \frac{\text{The generation of entropy as a consequence of heat transmission}}{\text{Aggregate generation of entropy}}.$$

BJ can be restated by employing (14) in the subsequent fashion:

$$BJ = \frac{(S_4 + Rd)\alpha_1 \theta'^2}{\alpha_1 (S_4 + Rd)\theta'^2 + \frac{Br}{S_2}\left(f''^2 + \alpha^2 g''^2\right) + S_3 Br\left(f'^2 + \alpha^2 g'^2\right)Mn}.$$

Results and discussion

Equations (7)–(9) are solved using the MATLAB inherent function bvp4c under conditions (10). In order to reduce the truncation error, we execute the grid independence test and record the results in Table 2. We calculate the Nusselt number with various mesh sizes to test the grid independence of the current code. In this research, we discover that a grid size of 200 guarantees the grid-independent approach.

Profiles of temperature and velocity

According to Figure 2(a), an increase in the Eckert number results in a higher fluid temperature. When the Eckert number goes up, it indicates an upsurge in the fluid's kinetic energy compared to its thermal energy. The rise in kinetic energy directly impacts the temperature of the fluid. Fluid temperature increases as the radiation parameter rises, as depicted in Figure 2(b). Thermal radiation is primarily about transferring heat energy through electromagnetic waves. When the thermal radiation parameter rises, it suggests that the fluid particles are absorbing more thermal radiation and act as an extra heat source for the fluid, resulting in an increase in temperature.

Table 2. Comparative analysis of Nusselt number across varying grid resolutions, at $Pr = 6.2$, $Ecn = 0.01$, $Mn = 1$, $Rd = 0.1$, $Bi = 0.3$, $\Gamma = 0.1$.

		Nusselt number		
Ecn	Mesh size	100	200	300
0		0.302864363	0.302864362	0.302864362
0.1		0.239590403	0.239590402	0.239590402
0.2		0.161952378	0.161952378	0.161952378

Figure 2(c) indicates that an upsurge in Γ result in a fall in the temperature of the fluid. The thermal relaxation parameter represents the time it takes for a fluid element to reach thermal equilibrium with its surroundings. When this parameter is increased, the heat transfer process amongst the fluid and its environment becomes slower. This delay in heat transfer allows the fluid to retain its thermal energy for a longer period, resulting in a lower overall temperature. Figure 2(d) illustrates the correlation among ϕ_T and the temperature of the fluid. Nanoparticles generally have higher thermal conductivity compared to the base fluid. This increased thermal conductivity allows the fluid to absorb and conduct heat more efficiently. As depicted in Figure 3(a) and (b), augmenting the magnetic field parameter results in a reduction of the fluid velocity. This is due to a force called the Lorentz force, which acts on any electrically conductive fluid that is moving through a magnetic field. In a fluid, the Lorentz force acts as an opposing force since it is opposed to both the fluid's velocity and the magnetic field. Raising the value of ϕ_T causes the fluid's velocity to decline, as can be seen in Figure 3(c) and (d). Note that the increase in the volume fraction of nanoparticles leads to higher viscosity and enhanced thermal interactions within the fluid. These factors combine to decrease the overall fluid velocity.

Irreversibility analysis

Figure 4(a) and (b) shows that as ϕ_T increases, the Bejan number rises and the same declines with the rise in Br. As Brinkman number increases, it signifies more internal heat

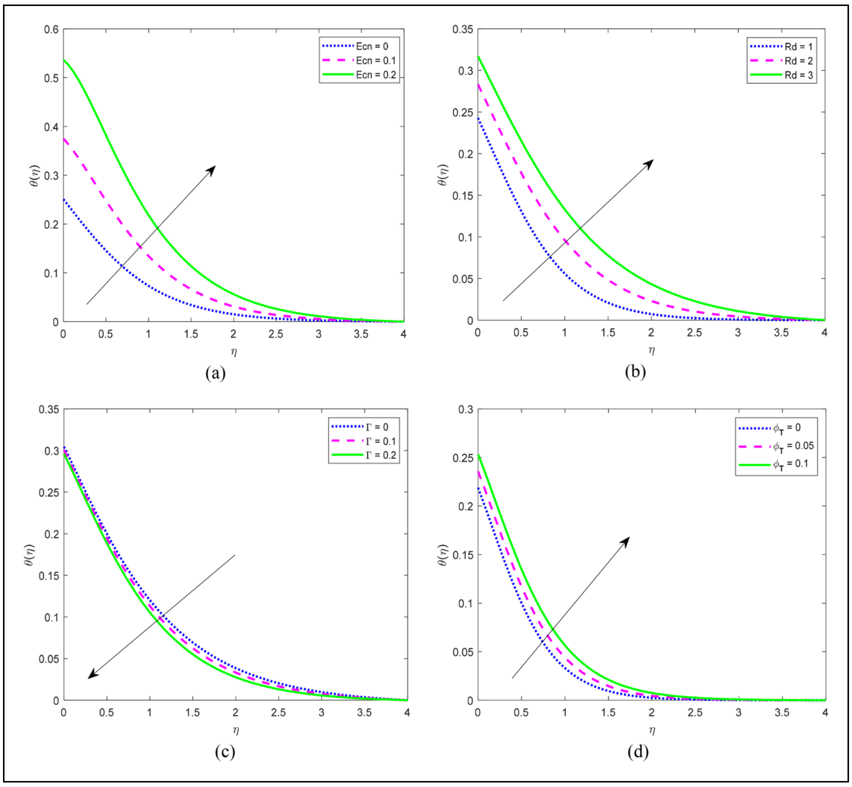

Figure 2. (a) Scenario in which $\theta(\eta)$ is affected by *Ecn*. (b) Scenario in which $\theta(\eta)$ is affected by *Rd*. (c) Scenario in which $\theta(\eta)$ is affected by Γ. (d) Scenario in which $\theta(\eta)$ is affected by ϕ_T.

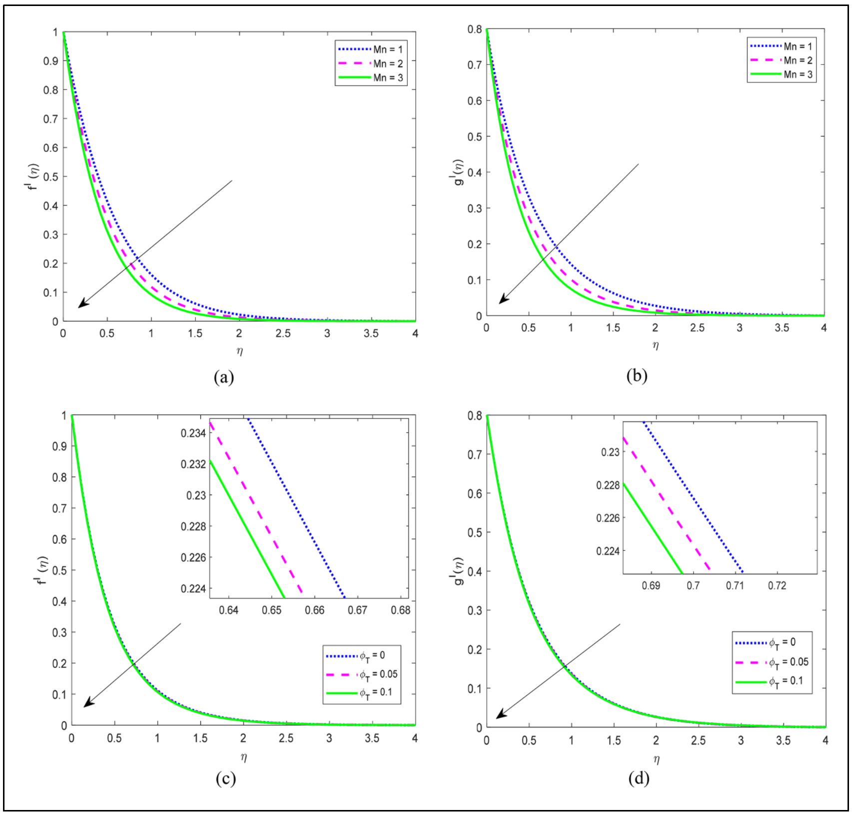

Figure 3. (a) Scenario in which$f'(\eta)$ is affected by *Mn* . (b) Scenario in which $g'(\eta)$ is affected by *Mn*. (c) Scenario in which $f'(\eta)$ is affected by ϕ_T. (d) Scenario in which $g'(\eta)$ is affected by ϕ_T.

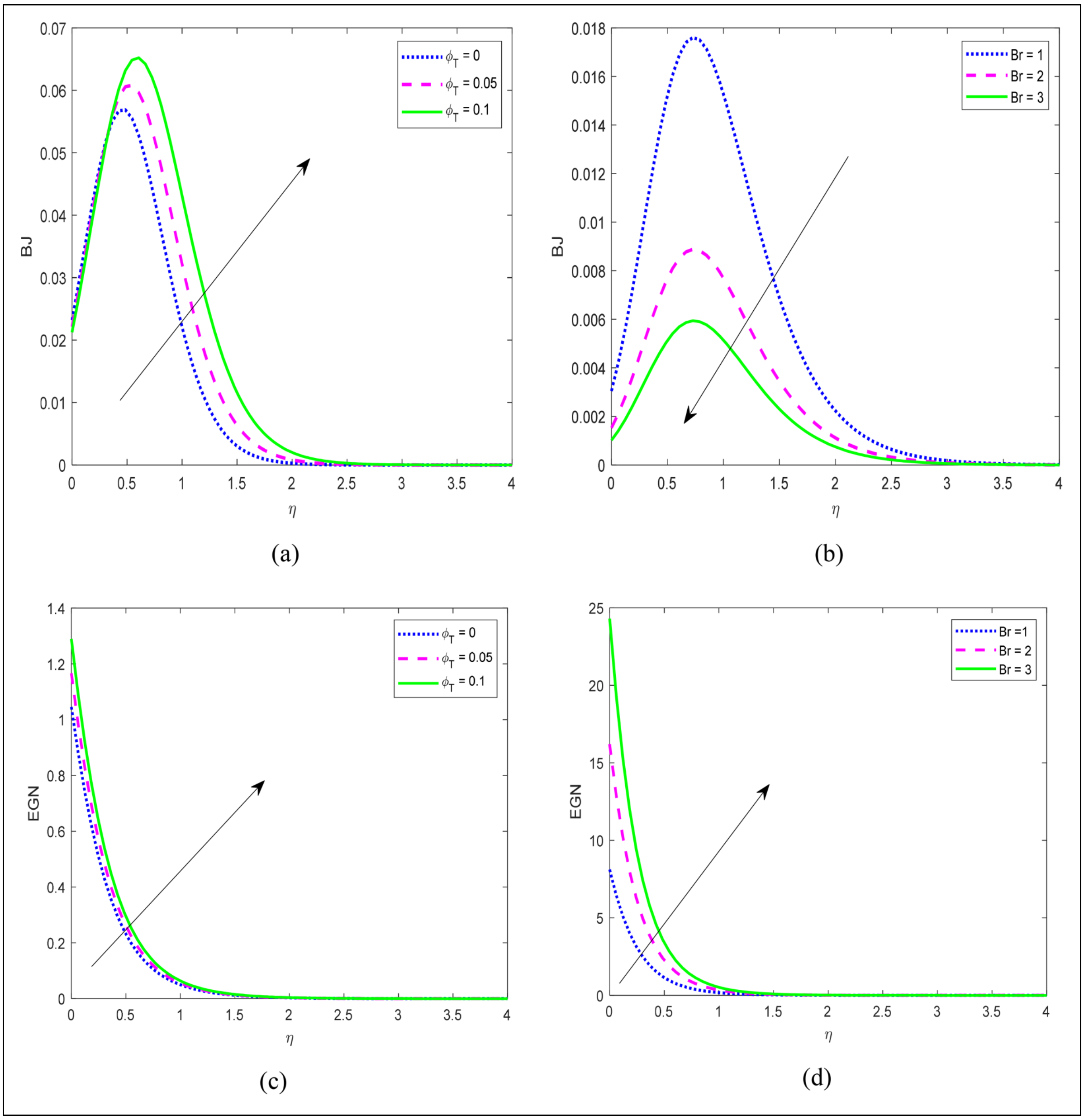

Figure 4. (A) Scenario in which *BJ* is affected by ϕ_T (b) scenario in which *BJ* is affected by *Br*. (c) Scenario in which *EGN* is affected by ϕ_T. (d) Scenario in which *EGN* is affected by *Br*.

generation due to viscous friction within the fluid. The additional heat generation due to friction contributes to the total entropy generation. This reduces the relative contribution of thermal conduction to entropy generation. Consequently, the Bejan number decreases as viscous dissipation becomes more significant. As can be seen in Figure 4(c) and (d), higher *Br* and ϕ_T values lead to a greater rate of entropy generation. Note that the increased volume fraction of nanoparticles leads to higher viscosity, frictional losses, and the creation of microscopic thermal gradients within the fluid. These factors combine to significantly increase the entropy generation in the flow.

Nusselt number and friction factor profiles

Figure 5(a)–(d) demonstrates that an increase in *Mn* and ϕ_T results in a decline of the skin friction factors (i.e., Cf_0 and Cf_1) and the friction factor declines at a rate of 0.62179204 (in case of Cf_0) and 0.621791816 (in case of Cf_1) respectively at $0 \le Mn \le 3.5$. Note that, with a rising magnetic field parameter, the opposing Lorentz force reduces the velocity of the fluid particles near the sheet, leading to a decrease in the skin friction coefficient. Figure 6(a)–(c) demonstrate the facts that there is a reduction in the Nusselt number with the rise in *Ecn* and the same rises with the rise in Γ and *Rd*. It is found that the Nusselt number declines at a rate of 0.825178645 when $0 \le Ecn \le 0.3$. As the Eckert number increases due to higher kinetic energy, more internal heat generation occurs within the fluid due to friction. When the fluid itself gets hotter due to internal friction (higher Eckert number), the temperature difference between the hot wall and the fluid reduces. This weakens the heat transfer process from the wall to the fluid, resulting in a lower

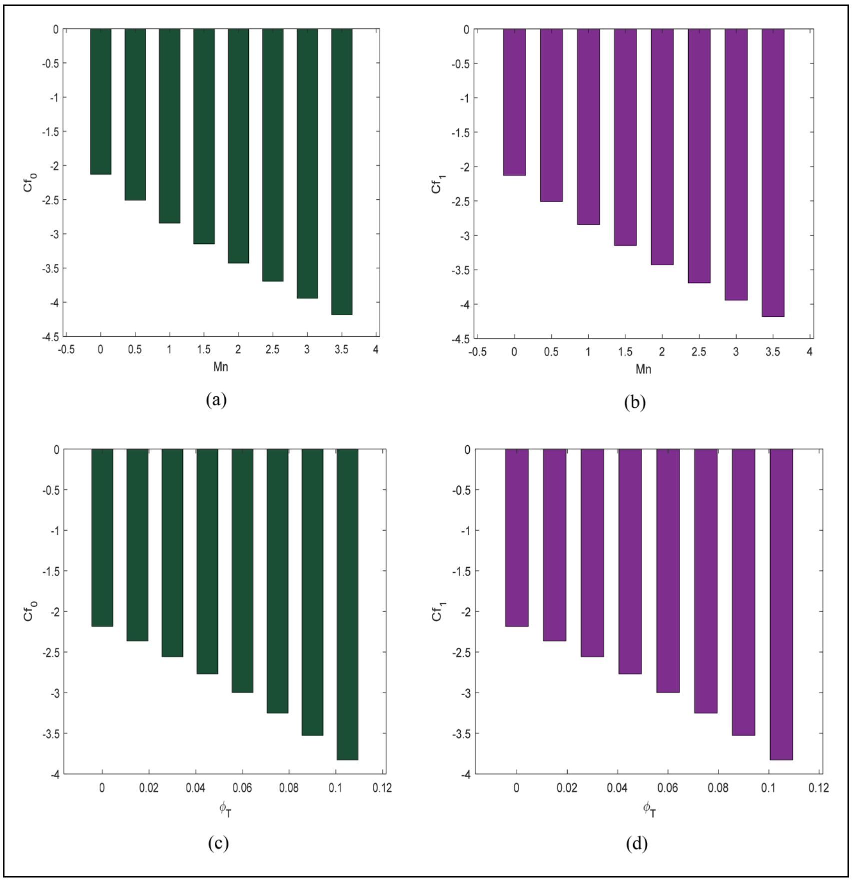

Figure 5. (a) Scenario in which Cf_0 is affected by Mn. (b) Scenario in which Cf_1 is affected by Mn. (c) Scenario in which Cf_0 is affected by ϕ_T. (d) Scenario in which Cf_1 is affected by ϕ_T.

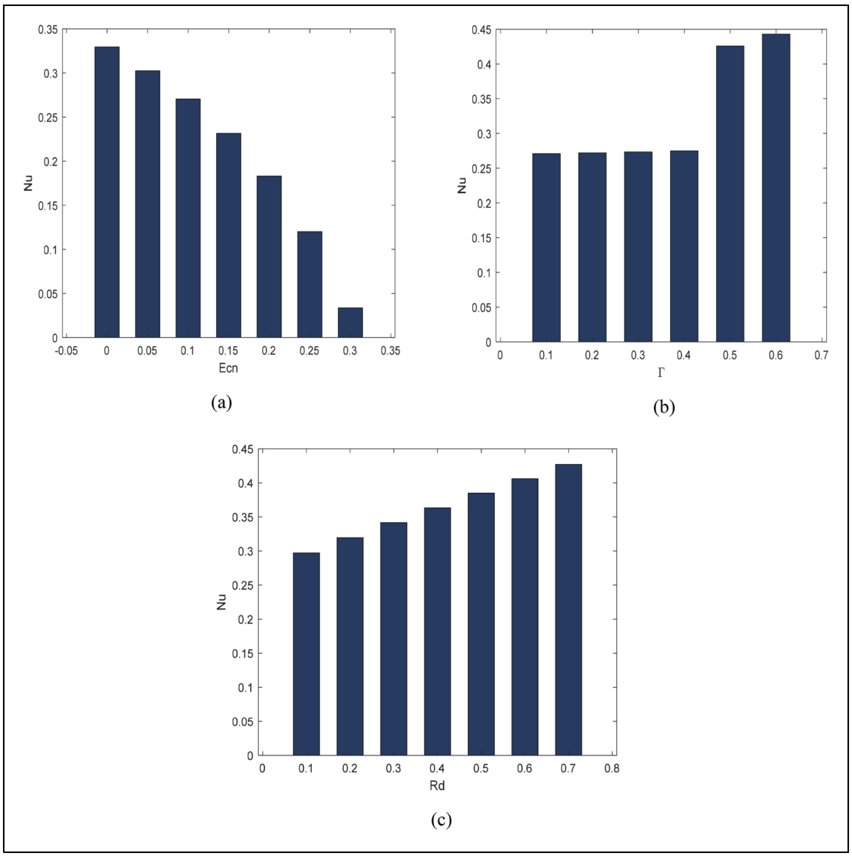

Figure 6. (a) Scenario in which Nu is affected by Ecn. (b) Scenario in which Nu is affected by Γ. (c) Scenario in which Nu is affected by Rd.

Table 3. Validation of current findings by comparing them to those of past studies conducted under more specialized settings such as $\phi_C = \phi_M = \phi_T = 0$.

	$-\theta'(0)$		
Pr	Nadeem and Hussain[46]	Gupta and Gupta[47]	Present results
0.2	0.169	0.16831182	0.168761064
0.7	0.454	0.45423652	0.454447363
2.0	0.911	0.91148310	0.911352811

Nusselt number. When the thermal radiation parameter increases, it means more radiative energy gets absorbed by the fluid particles. This absorbed energy directly translates to a rise in the internal energy and consequently, the temperature of the fluid particles. Due to this temperature difference between the hot surface and the cooler fluid, heat transfer by convection becomes more vigorous. As a result, the Nusselt number, which reflects the efficiency of convective heat transfer, increases.

Validation

To confirm the validity of our results, we compare them to previously published data under specific conditions. An acceptable agreement is observed (refer to Table 3).

Conclusions

Fluid flow over a bidirectional stretching sheet holds significance in various scientific and industrial applications due to its ability to model situations involving continuous deformation and heat transfer. This work theoretically investigates the ternary hybrid nanofluid (composed of water, titanium dioxide, cobalt ferrite, and magnesium oxide) flow through a bidirectional sheet. The analysis takes into account ohmic heating, Cattaneo–Christov heat flux, irreversibility, thermal radiation, and viscous dissipation. The findings of the current investigation can be succinctly summarized thereby:

- Higher Eckert number and thermal radiation parameter lead to increased temperature.
- Increased magnetic field parameter and volume fraction of nanoparticles reduce fluid velocity.
- Rising Brinkman number increases entropy generation and decreases the Bejan number. Higher nanoparticle volume fraction also increases entropy generation
- The magnetic field parameter and volume fraction of nanoparticles decrease the skin friction coefficient due to the Lorentz force reducing velocity near the sheet.
- A higher Eckert number reduces the Nusselt number. Conversely, increasing thermal radiation and thermal relaxation parameters enhance the Nusselt number.
- Higher thermal relaxation time allows the fluid to retain heat, resulting in a lower temperature.
- It is revealed that there is a reduction in the Nusselt number with the rise in Eckert number (Ecn) and the same number declines at a rate of 0.8252 when $0 \leq Ecn \leq 0.3$.
- It is noticed that the skin friction coefficient declines at a rate of 0.62179204 (in case of $x-$direction) and 0.621791816 (in case of $y-$direction), respectively, when the values of magnetic field parameter lie between 0 and 3.5.

The study suggests that we can control a fluid's behavior by adjusting these parameters. Nanoparticles offer a trade-off between heat transfer and velocity due to their impact on viscosity. Understanding these relationships allows engineers to optimize heat transfer efficiency and minimize energy losses in various applications.

Data availability

Data sharing not applicable to this article as no data sets were generated or analyzed during the current study.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

ORCID iDs

KS Srinivasa Babu https://orcid.org/0000-0002-9292-8214
M Jayachandra Babu https://orcid.org/0000-0001-7645-6301

References

1. Medebber MA, Aissa A, Slimani MEA, et al. Numerical study of natural convection in vertical cylindrical annular enclosure filled with Cu-water nanofluid under magnetic fields. *Defect Diffus Forum* 2019; 392: 123–137.
2. Radouane F, Abderrahmane A, Mebarek-Oudina F, et al. Magneto-free convectiveof hybrid nanofluid inside non-darcy porous enclosure containing an adiabatic rotating cylinder. *Energy Sources Part A* 2020: 1–16. doi:10.1080/15567036.2020.1856235.
3. Jamshed W, Eid MR, Aissa A, et al. Partial velocity slip effect on working magneto non-Newtonian nanofluids flow in solar collectors subject to change viscosity and thermal conductivity with temperature. *PLos One* 2021; 16: e0259881.
4. Wakif A, Abderrahmane A, Guedri K, et al. Importance of exponentially falling variability in heat generation on chemically reactive von Kármán nanofluid flows subjected to a radial magnetic field and controlled locally by zero mass flux and convective heating conditions: a differential quadrature analysis. *Front Phys* 2022; 10: 988275.
5. Shamshuddin MD, Rajput GR, Jamshed W, et al. MHD Bioconvection microorganism nanofluid driven by a stretchable plate through porous media with an induced heat source. *Waves Random Complex MEdia* 2022: 1–25. doi:10.1080/17455030.2022.2126024.

6. Choudhary S, Mehta R, Alessa N, et al. Thermal analysis on kerosene oil-based two groups of ternary hybrid nanoparticles (CNT-Gr-Fe_3O_4 and MgO-Cu-Au) mix flow over a bidirectional stretching sheet: a comparative approach. *J Engg* 2023; 2023: 1–20.
7. Pavithra KM, Srilatha P, Hanumagowda BN, et al. A free convective two-phase flow of optically thick radiative ternary hybrid nanofluid in an inclined symmetrical channel through a porous medium. *Symmetry (Basel)* 2023; 15: 1615.
8. Nagaraja KV, Khan U, Madhukesh JK, et al. Heat and mass transfer analysis of assisting and opposing radiative flow conveying ternary hybrid nanofluid over an exponentially stretching surface. *Sci Rep* 2023; 13: 14795.
9. Sarfraz M, Khan M, Al-Zubaidi A, et al. Tribology-informed analysis of convective energy transfer in ternary hybrid nanofluids on inclined porous surfaces. *Tribol Int* 2023; 188: 108860.
10. Yaseen M, Rawat SK, Shah NA, et al. Ternary hybrid nanofluid flow containing gyrotactic microorganisms over three different geometries with Cattaneo–Christov model. *Mathematics* 2023; 11: 1237.
11. Prakash J, Tripathi D and Bég OA. Computation of EMHD ternary hybrid non-Newtonian nanofluid over a wedge embedded in a Darcy-Forchheimer porous medium with zeta potential and wall suction/injection effects. *Int J Ambient Energy* 2023; 44: 2155–2169.
12. Mahboobtosi M, Hosseinzadeh K and Ganji DD. Entropy generation analysis and hydrothermal optimization of ternary hybrid nanofluid flow suspended in polymer over curved stretching surface. *Int J Thermofluids* 2023; 20: 100507.
13. Hasnain J and Abid N. Numerical investigation for thermal growth in water and engine oil-based ternary nanofluid using three different shaped nanoparticles over a linear and nonlinear stretching sheet. *Numer Heat Transfer Part A* 2023; 83: 1365–1376.
14. Madhukesh JK, Sarris IE, Prasannakumara BC, et al. Investigation of thermal performance of ternary hybrid nanofluid flow in a permeable inclined cylinder/plate. *Energies* 2023; 16: 2630.
15. Mahmood Z, Khan U, Saleem S, et al. Numerical analysis of ternary hybrid nanofluid flow over a stagnation region of stretching/shrinking curved surface with suction and lorentz force. *J Magn Magn Mater* 2023; 573: 170654.
16. Guedri K, Mahmood Z, Fadhl BM, et al. Mathematical analysis of nonlinear thermal radiation and nanoparticle aggregation on unsteady MHD flow of micropolar nanofluid over shrinking sheet. *Heliyon* 2023; 9: e14248.
17. Rafique K, Mahmood Z, Alqahtani AM, et al. Impacts of thermal radiation with nanoparticle aggregation and variable viscosity on unsteady bidirectional rotating stagnation point flow of nanofluid. *Mater Today Commun* 2023; 36: 106735.
18. Mukhtar T, Jamshed W, Aziz A, et al. Computational investigation of heat transfer in a flow subjected to magnetohydrodynamic of Maxwell nanofluid over a stretched flat sheet with thermal radiation. *Numer Methods Partial Differ Equ* 2023; 39: 3499–3519.
19. Salawu SO, Obalalu AM and Shamshuddin MD. Nonlinear solar thermal radiation efficiency and energy optimization for magnetized hybrid Prandtl–Eyring nanoliquid in aircraft. *Arab J Sci Eng* 2023; 48: 3061–3072.
20. Ramesh GK, Madhukesh JK, Shah NA, et al. Flow of hybrid CNTs past a rotating sphere subjected to thermal radiation and thermophoretic particle deposition. *Alexandria Eng J* 2023; 64: 969–979.
21. Suresh Kumar Y, Hussain S, Raghunath K, et al. Numerical analysis of magnetohydrodynamics Casson nanofluid flow with activation energy, Hall current and thermal radiation. *Sci Rep* 2023; 13: 4021.
22. Dharmaiah G, Mebarek-Oudina F, Sreenivasa Kumar M, et al. Nuclear reactor application on Jeffrey fluid flow with Falkner-Skan factor, Brownian and thermophoresis, non linear thermal radiation impacts past a wedge. *J Indian Chem Soc* 2023; 100: 100907.
23. Lund LA, Asghar A, Rasool G, et al. Magnetized casson SA-hybrid nanofluid flow over a permeable moving surface with thermal radiation and Joule heating effect. *Case Stud Therm Eng* 2023; 50: 103510.
24. Fourier JBJ. *Théorie analytique de la chaleur*. Paris: Gauthier-Villars et fils, 1888.
25. Cattaneo C. Sulla conduzione del calore. *Atti Sem Mat Fis Univ Modena* 1948; 3: 83–101.
26. Christov CI. On frame indifferent formulation of the Maxwell–Cattaneo model of finite-speed heat conduction. *Mech Res Commun* 2009; 36: 481–486.
27. Ali U, Irfan M, Rehman KU, et al. On the Cattaneo–Christov heat flux theory for mixed convection flow due to the rotating disk with slip effects. *Waves Random Complex MEdia* 2022: 1–15. doi:10.1080/17455030.2022.2056659.
28. Khattak S, Ahmed M, Abrar MN, et al. Numerical simulation of Cattaneo–Christov heat flux model in a porous media past a stretching sheet. *Waves Random Complex MEdia* 2022: 1–20. doi:10.1080/17455030.2022.2030503.
29. Qureshi MA. Irreversibility analysis of electromagnetic hybrid nanofluid for Cattaneo–Christov heat flux model using finite element approach. *Sci Rep* 4288; 13: 2023.
30. Rehman S, Ali H, Trabelsi Y, et al. A renovated Jaffrey-Hamel flow problem and new scaling statistics for heat, mass fluxes with Cattaneo–Christov heat flux model. *Case Stud Therm Eng* 2023; 43: 102787.
31. Punith Gowda R, Naveen Kumar R, Kumar R, et al. Three-dimensional coupled flow and heat transfer in non-Newtonian magnetic nanofluid: an application of Cattaneo-Christov heat flux model. *J Magn Magn Mater* 2023; 567: 170329.
32. Fatima N, Hasnain J, Sanaullah et al. Aspects of Cattaneo-Christov heat flux in nonlinear radiative ternary, hybrid, and single mass diffusion past stretching surface; A comparative study. *Case Stud Therm Eng* 2023; 43: 102776.
33. Saraswathy M, Prakash D, Muthtamilselvan M, et al. Theoretical study on bio-convection of micropolar fluid with an exploration of Cattaneo–Christov heat flux theory. *Int J Mod Phys B* 2024; 38: 2450016.
34. Elshehabey HM, Aly AM, Lee S-W, et al. Integrating artificial intelligence with numerical simulations of Cattaneo-Christov heat flux on thermosolutal convection of nano-enhanced phase change materials in Bézier-annulus. *J Energy Storage* 2024; 82: 110496.
35. Rehman MIU, Chen H, Hamid A, et al. Analysis of Cattaneo–Christov heat flux and thermal radiation on Darcy–Forchheimer flow of Reiner–Philippoff fluid. *Int J Mod Phys B* 2024; 38: 2450046.
36. Manjunatha S and Gireesha BJ. Effects of variable viscosity and thermal conductivity on MHD flow and heat transfer of a dusty fluid. *Ain Shams Eng J* 2016; 7: 505–515.

37. Umavathi JC, Chamkha AJ and Mohiuddin S. Combined effect of variable viscosity and thermal conductivity on free convection flow of a viscous fluid in a vertical channel. *Int J Numer Methods Heat Fluid Flow* 2016; 26: 18–39.
38. Ajibade AO and Tafida MK. The combined effect of variable viscosity and variable thermal conductivity on natural convection couette flow. *Int J Thermofluids* 2020; 5: 100036.
39. Aziz A, Hussain ST and Sadiq A. Mathematical model for a thermal cooling system with variable viscosity and thermal conductivity over a rotating disk. *Case Stud Therm Eng* 2023; 52: 103664.
40. Porgar S, Oztop HF and Salehfekr S. "A comprehensive review on thermal conductivity and viscosity of nanofluids and their application in heat exchangers". *J Mol Liq* 2023; 386: 122213.
41. Mandal G and Pal D. Dual solutions of radiative Ag-MoS_2/water hybrid nanofluid flow with variable viscosity and variable thermal conductivity along an exponentially shrinking permeable Riga surface: stability and entropy generation analysis. *Int J Model Simul* 2023: 1–26. doi:10.1080/02286203.2023.2171656.
42. Azam M, Khan WA and Nayak MK. Three-dimensional flow of cross nanofluid over bidirectional moving surface in Darcy-Forchheimer medium with Cattaneo-Christov heat flux. *Case Stud Therm Eng* 2023; 49: 103250.
43. Ali B, Thumma T, Habib D, et al. Finite element analysis on transient MHD 3D rotating flow of Maxwell and tangent hyperbolic nanofluid past a bidirectional stretching sheet with Cattaneo Christov heat flux model. *Therm Sci Eng Prog* 2022; 28: 101089.
44. Alharbi KAM, Ahmed AES, Ould Sidi M, et al. Computational valuation of Darcy ternary-hybrid nanofluid flow across an extending cylinder with induction effects. *Micromachines (Basel)* 2022; 13: 88.
45. Liu I-C and Andersson HI. Heat transfer over a bidirectional stretching sheet with variable thermal conditions. *Int J Heat Mass Transfer* 2008; 51: 4018–4024.
46. Nadeem S and Hussain ST. Flow and heat transfer analysis of Williamson nanofluid. *Appl Nanosci* 2014; 4: 1005–1012.
47. Gupta S and Gupta S. MHD three-dimensional flow of Oldroyd-B nanofluid over a bidirectional stretching sheet: DTM-Padé solution. *Nonlinear Eng* 2019; 8: 744–754.

Nomenclature

σ	Electrical conductivity $[Sm^{-1}]$
Pr	Prandtl number
ϕ_C	Volume fraction of $CoFe_2O_4$
Γ	Thermal relaxation parameter
ρ	Density $[kgm^{-3}]$
β_T, β_{1T}	Thermal expansion coefficients $[K^{-1}]$
α_1	Temperature difference parameter
κ	Porosity parameter
δ	Inverse porosity parameter
Γ^*	Thermal relaxation time of heat Flux $[s]$
C_p	Specific heat $[Jkg^{-1}K^{-1}]$
h_f	Convection heat transfer coefficient $[Wm^{-2}K^{-1}]$
Mn	Magnetic field Parameter
α, λ	Stretching ratio parameters
η	Similarity variable
ϕ_M	Volume fraction of MgO
$\sigma*$	Stefan–Boltzmann constant $[Wm^{-2}K^{-4}]$
ϕ_T	Volume fraction of TiO_2
t	Temperature ratio parameter
Br	Brinkman number
τ, τ_1	Convection parameters (mixed and nonlinear)
Re_x	Reynolds number
Ecn	Eckert number
Bi	Biot number
q_w	Wall heat flux $[Wm^{-2}]$
Rd	Thermal radiation parameter
k	Thermal conductivity $[Wm^{-1}K^{-1}]$
k^*	Mean absorption coefficient $[m^2mol^{-1}]$
B_0	Strength of magnetic field $[T]$